

Yunfan Xie (Ryan)

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Education

The Hong Kong University of Science and Technology

Hong Kong

BEng in Computer Engineering, Minor in Robotics, Extended in AI

2024 – Present

- CGA: 3.816 / 4.3, Major CGA: 3.900 / 4.3
- Relevant Coursework: Data Structures & Algorithms, Embedded Systems, Robotics Foundations, Computer Architecture, Control Systems, C++ Programming

Skills

Programming: C, C++, Python, MATLAB, JavaScript, HTML, CSS

Embedded & Robotics: STM32, Motor Control, ROS, Sensor Integration, Circuit Design

Tools: Git, Linux, SolidWorks, VS Code, LaTeX, KiCad

Spoken Languages: English (fluent), Mandarin (native), Cantonese (intermediate)

Project Experience

Terralix Smart Ski Goggles

2026 – Present

Software Developer

- Deployed models on RK3588 board for edge AI computing.
- Exploring LiDAR-based snow terrain recognition algorithm.

HKUST RoboMaster Team ENTERPRIZE

2025 – Present

Embedded Development

- Led the embedded development for the Sentry robot, creating the core control logic for autonomous decision-making. Developed control algorithms to improve navigation accuracy and target tracking.
- Improved IMU fusion and AHRS algorithms, decreased the response latency by 60%, and increased the control precision by 30%.
- Worked with mechanical and hardware teams to integrate sensors and motors, ensuring smooth system operation.

USTLeaf Project (ustleaf.cloud)

2024 – 2025

Full-Stack Developer

- Built and deployed ustleaf.cloud with peers, a website that catalogs plant species at HKUST with detailed photos and information.
- Added community features that allow users to upload their own photos, post comments, and like posts.

Awards & Achievements

RoboMaster

2026

- RMUL 2026 3V3 Sentry Second Prize
- RMUC 2026 First Prize

HKUST Academic Achievements

2024 – Present

- Dean's List (Top 1%)
- University Admission Scholarship
- Continuing UG Academic Scholarship

National Olympiad in Informatics in Provinces (NOIP)

2019

- NOIP First Prize for outstanding performance in algorithmic problem-solving in C++.

Interests & Objectives

Passionate about software programming, embedded systems and algorithm. Seeking research opportunities or internships.